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Project Title:	Dose-rate effects of protons on the induction of genomic instability in vivo		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	11794-8691	**Congressional District:**	1
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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Exposure to solar particle events (SPEs) and the heavy ions (HZE) found in space are the greatest hazard to astronauts venturing beyond Earth. However, little is known about their dose/dose-rate effects. During a journey to Mars, every cell in the body will be hit by a proton once every three days, and by a HZE ion approximately once a month. One of the major concerns associated with the exposure of the astronauts to protons during space flights is cancer induction. It is therefore important to protect astronauts in space environments, which will require improvements in our knowledge of radiation-induced cellular and molecular changes that impact the health risks of astronauts. While biological effects of HZE particles found in space (i.e. 56Fe ions) have been investigated extensively, little is known about the in vivo effects of SPEs, especially protons, which are the most abundant component of the SPEs. The work outlined for this in vivo project is designed to fill in this knowledge gap. For data about radiation-induced damage to be most useful in reducing

Task Description:	the uncertainties in the assessment of health risks, they must be established at proper doses/dose rates in in vivo systems because in vitro systems normally do not fully mimic in vivo situations. This study is designed to determine the dose-rate effects of protons (100 MeV/amu) on the in vivo induction of genomic instability (expressed as delayed chromosomal damage). This approach is highly relevant for assessing cancer risks; however, it has not been applied with proton exposure. We will characterize the type, the frequency, and the distribution of initial and subsequent chromosome aberrations, and the expression of genomic instability induced in vivo in bone marrow cells after exposure of mice to 100MeV/amu protons at the dose and dose rates encountered in space. A novel genome-wide multicolor fluorescence in situ hybridization (mFISH) of mouse chromosomes will be used to detect and characterize chromosome aberrations in metaphase cells prepared from bone marrow cells of exposed mice collected at different times (up to 6 months) post-irradiation. To afford a highly reliable risk assessment, a much improved understanding of the molecular basis for dose-rate effects of protons is needed. Therefore, this research project is also designed to determine the potential molecular basis for the dose-rate dependencies of proton-induced genomic instability in vivo. To achieve this goal, we will carefully measure levels of activated nuclear factor-kappaB (NF-kappaB), a highly important transcription factor (a protein that controls the transcription of the genes), and the expression of selected NF-kappaB target proteins involved in cell protection (i.e. manganese superoxide dismutase and catalase) and inflammatory response in bone marrow cells collected at early times after irradiation. This will be coupled with the quantitation of two protein kinases, DNA-dependent protein kinase (DNA-PK) and the ataxia telangiectasia mutated protein kinase (ATM). We anticipate that these kinases will be crucial for NF-kappaB activation after irradiation, leading to the activation of proteins involved in inflammatory response and cell survival. Thus, activation of NF-kappaB may play a pivotal role in the maintenance of cells carrying genomic instability. This study directly addresses a key question that NASA is trying to answer regarding the cancer risk associated with space exploration. The results from this research should also substantially improve the understanding of the potential mechanisms for the dose-rate effects of exposure to protons.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	New exploration of Moon and Mars by the US-NASA will undoubtedly happen in the very near future. It is therefore important to learn more about potential health effects of space radiation before the manned missions to these other worlds become a routine adventure. Such information should help reduce the uncertainties associated with risk assessment for human space exploration. Upon completion of the in vivo work proposed in this study, our data will definitely provide pivotal information that should lead to a better understanding of dose-rate effects and of mechanisms by which proton-induced aberrations can be potentially involved in the induction of genomic instability and cancer. The results also would facilitate determining whether there are unique biological effects associated with in vivo exposure to protons. Moreover, the mFISH technology used in the research project has the power to facilitate the discovery of genetic changes in other cancers that are unrelated to ionizing radiation. Thus, this research program is broadly relevant and a valuable resource to the many other investigators throughout the scientific community. In addition, this study is important for understanding risks associated with space travel as well as developing advanced cancer therapies for Earth because it will couple molecular mechanisms with quantitative data from a robust, highly sensitive technique.
Task Progress:	This project was initiated in October 2007. Therefore, the content in this Task Book Report represents the accomplishments of the past 7 months. There was no proton-beam time available for us during the Spring-2008 run. We, however, obtained the proton-beam time in the Summer-2008 run (NSRL-08) on June 14, 2008. We exposed 10 mice to 1.0 Gy of 100 MeV/amu protons, at the dose rate of 1 cGy/min. Another 10 mice received 0 Gy of 100 MeV/amu protons and served as sham controls. At 3 hr post-irradiation, we collected bone marrow cells from 5 sham controls and 5 exposed mice for further analyses. This harvest time was performed at the Brookhaven Laborarory Animal Care Facility (BLAF). The rest of mice were brought to Stony Brook University on June 19, 2008. Sample collections from this group of mice will be done at 6 months post-irradiation. In November of 2007, we exposed 2 groups of mice (10 mice in each) to 0 and 1.0 Gy of 137Cs gamma rays (a reference radiation, at the dose rate of 1 cGy/min), respectively, using the gamma source from the Controlled Environment Radiation Facility (CERF) of the Biology Department, BNL. Similar to the proton-exposed group, mice receiving 0 Gy of radiation served as sham controls. At 1 month post-irradiation, BM cells were collected from 5 sham controls and 5 exposed mice for further analysis. Sample collection from the rest of the mice was done at 6 months post-irradiation. Using the mFISH method, the analyses of chromosome aberrations are in progress. To date, we have data on chromosome aberrations detected in bone marrow cells collected at 1 month after exposure of BALB/cJ mice to 0 or 1.0 Gy of 137Cs gamma rays (delivered at 1 cGy/min). We detected several types of aberrations in bone marrow cells collected from mice exposed to 1.0 Gy of 137Cs gamma rays. These were chromosomal exchanges and breaks (both chromatid- and chromosome-types). Chromosomal exchanges included translocations [Robertsonian (RT) or centric fusion, and incomplete types]. In general, we found no evidence of a non-random involvement of specific chromosomes in any type of aberrations observed in control mice or mice exposed to 1.0 Gy of 137Cs gamma rays, delivered at 1 cGy/min. Our data show a statistically significant increase ($p<0.01$, student's t-test) in the frequency of CAs, including abnormal cells, after exposure of mice to 1.0 Gy of 137Cs gamma rays, as compared to that found in control mice. The analyses of mFISH metaphase chromosomes in the rest of the mice from this harvest time and from the other groups of mice are in progress. Although the mFISH method is robust and sensitive, it is very time consuming. Due to a very large number of metaphase cells to be scored, the evaluation of genome-wide chromosome aberrations by mFISH will be the major effort for each year of the study. We work diligently to generate data on the potential induction of genomic instability (measured by late occurring CAs) by protons and gamma rays for NASA. Our study is important because it has been suggested that an elevation of genomic instability correlates with an increased risk of cancer. During the next years, studies will focus on the characterization of the type and distribution of initial and the subsequent loss of chromosomal damage, the expression of genomic instability induced in vivo in bone marrow cells. Dose and time response relationships will be established for the induction of genomic instability. Moreover, levels of activated NF-kappaB and expression of selected NF-kappaB target proteins, and kinase activities of DNA-PK and ATM

	will be obtained from bone marrow cells collected at early harvest times. It is anticipated that studies of the effects of protons administered at the dose rate of 1 cGy/min of protons and 137Cs gamma rays will be completed in year 2 of the study; whereas those for a lower dose-rate (0.5 cGy/min) will be done in years 3 and 4 of the study.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Rithidech K, Honikel L, Simon SR, Spitz D, Whorton EB, Anderson CW. "Dose-rate Effects of Protons on the Induction of Genomic Instability in vivo." 19th Annual NASA Space Radiation Health Investigators' Workshop, June 30-July 2, 2008, Philadelphia, PA. 19th Annual NASA Space Radiation Health Investigators' Workshop, June 30-July 2, 2008, Philadelphia, PA. , Jun-2008